

15 ДЗ - Профилирование запросов.

Собираем Dockerfile.

clickhouse-25.2.1.Dockerfile:

```
FROM clickhouse/clickhouse-server:25.2.1
MAINTAINER Maksim Kulikov <max.uoles@rambler.ru>

RUN apt-get update -y --fix-missing
RUN DEBIAN_FRONTEND=noninteractive apt-get -yq upgrade
RUN apt-get install nano mc python3 pip kafkacat -y
RUN pip install clickhouse_driver

EXPOSE 8123 9000

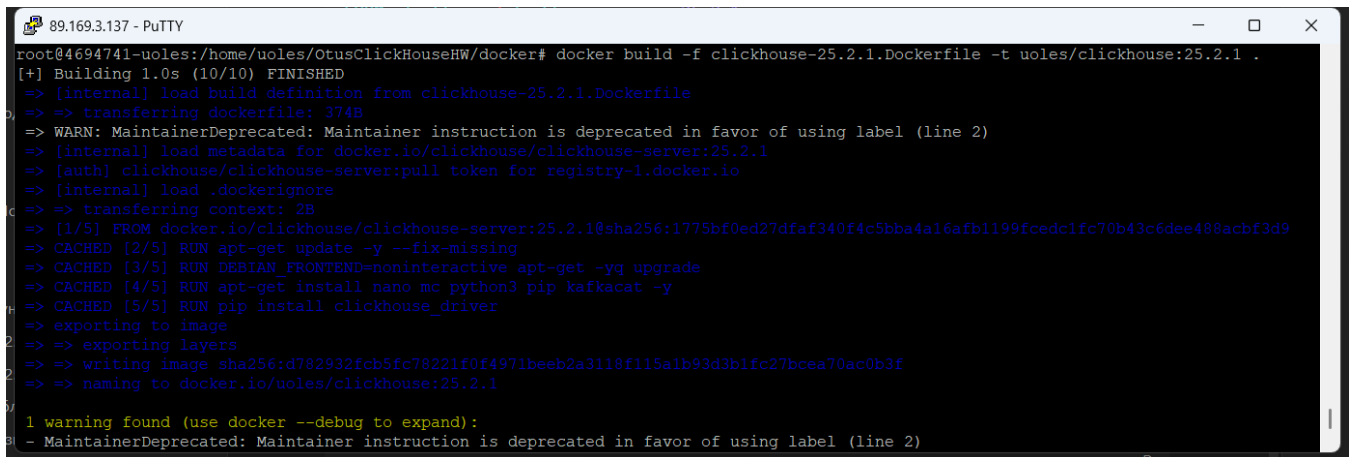
ENTRYPOINT ["/entrypoint.sh"]
```

Собираем образ командой:

```
docker build -f clickhouse-25.2.1.Dockerfile -t uoles/clickhouse:25.2.1 .
```

Запускаем:

```
docker run -d \
-p 18123:8123 \
-p 19000:9000 \
-e CLICKHOUSE_DB=my_database \
-e CLICKHOUSE_USER=username \
-e CLICKHOUSE_DEFAULT_ACCESS_MANAGEMENT=1 \
-e CLICKHOUSE_PASSWORD=password \
--name clickhouse-server \
--ulimit nofile=262144:262144 \
uoles/clickhouse:25.2.1
```



```
89.169.3.137 - PuTTY
root@4694741-uoles:/home/uoles/OtusClickHouseHW/docker# docker build -f clickhouse-25.2.1.Dockerfile -t uoles/clickhouse:25.2.1 .
[+] Building 1.0s (10/10) FINISHED
=> [internal] load build definition from clickhouse-25.2.1.Dockerfile
=> => transferring dockerfile: 374B
=> WARN: MaintainerDeprecated: Maintainer instruction is deprecated in favor of using label (line 2)
=> [internal] load metadata for docker.io/clickhouse/clickhouse-server:25.2.1
=> [auth] clickhouse/clickhouse-server:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/clickhouse/clickhouse-server:25.2.1@sha256:1775bf0ed27dfaf340f4c5bba4a16afb1199fcedc1fc70b43c6dee488acbf3d9
=> CACHED [2/5] RUN apt-get update -y --fix-missing
=> CACHED [3/5] RUN DEBIAN_FRONTEND=noninteractive apt-get -yq upgrade
=> CACHED [4/5] RUN apt-get install nano mc python3 pip kafkacat -y
=> CACHED [5/5] RUN pip install clickhouse_driver
=> exporting to image
=> => exporting layers
=> => writing image sha256:d782932fcb55fc78221f0f4971beeb2a3118f115a1b93d3b1fc27bcea70ac0b3f
=> => naming to docker.io/uoles/clickhouse:25.2.1

1 warning found (use docker --debug to expand):
- MaintainerDeprecated: Maintainer instruction is deprecated in favor of using label (line 2)
```

Создание таблицы и тестирование запросов.

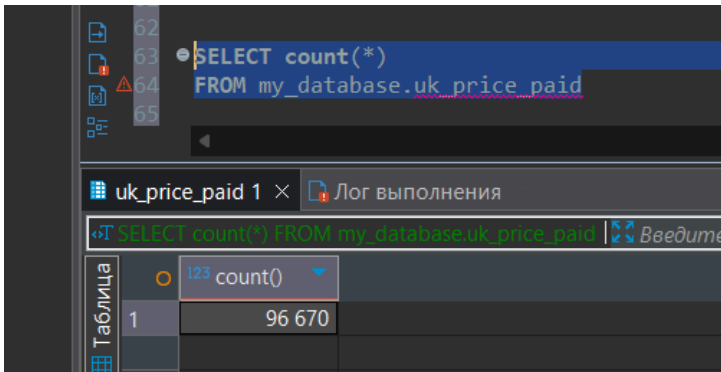
Использую датасет (ежемесячное обновление):

<https://clickhouse.com/docs/getting-started/example-datasets/uk-price-paid>

Создаем таблицу и заполняем данными:

```
CREATE TABLE my_database.uk_price_paid
(
    price UInt32,
    date Date,
    postcode1 LowCardinality(String),
    postcode2 LowCardinality(String),
    type Enum8('terraced' = 1, 'semi-detached' = 2, 'detached' = 3, 'flat' = 4, 'other' =
0),
    is_new UInt8,
    duration Enum8('freehold' = 1, 'leasehold' = 2, 'unknown' = 0),
    addr1 String,
    addr2 String,
    street LowCardinality(String),
    locality LowCardinality(String),
    town LowCardinality(String),
    district LowCardinality(String),
    county LowCardinality(String)
)
ENGINE = MergeTree
ORDER BY (postcode1, postcode2, addr1, addr2);

INSERT INTO my_database.uk_price_paid
SELECT
    toUInt32(price_string) AS price,
    parseDateTimeBestEffortUS(time) AS date,
    splitByChar(' ', postcode)[1] AS postcode1,
    splitByChar(' ', postcode)[2] AS postcode2,
    transform(a, ['T', 'S', 'D', 'F', 'O'], ['terraced', 'semi-detached', 'detached',
'flat', 'other']) AS type,
    b = 'Y' AS is_new,
    transform(c, ['F', 'L', 'U'], ['freehold', 'leasehold', 'unknown']) AS duration,
    addr1,
    addr2,
    street,
    locality,
    town,
    district,
    county
FROM url(
    'http://prod.publicdata.landregistry.gov.uk.s3-website-eu-west-1.amazonaws.com/pp-
monthly-update-new-version.csv',
    'CSV',
    'uuid_string String,
    price_string String,
    time String,
    postcode String,
    a String,
    b String,
    c String,
    addr1 String,
    addr2 String,
    street String,
    locality String,
    town String,
    district String,
    county String,
    d String,
    e String'
) SETTINGS max_http_get_redirects=10;
```



Проверяем использование индексов через лог.

Заходим в docker контейнер и подключаемся к clickhouse через clickhouse-client:

```
>> docker exec -it clickhouse-server clickhouse-client --user username --password password
```

Выставляем уровень логирования – trace:

```
>> SET send_logs_level = 'trace';
```

Проверяем запрос без primary_key поля:

```
SELECT *
FROM my_database.uk_price_paid
WHERE price = 14000

Query id: 09b46531-elae-418d-a429-c06640829ea3

[255c4e34d10] 2025.05.11 14:13:29.537506 [ 769 ] (09b46531-elae-418d-a429-c06640829ea3) <Debug> executeQuery: (from 127.0.0.1:60446, user: username) (query 1, line 1) select * from my_data
base.uk_price_paid where price = 14000; (stage: Complete)
[255c4e34d10] 2025.05.11 14:13:29.538070 [ 769 ] (09b46531-elae-418d-a429-c06640829ea3) <Trace> Planner: Query to stage Complete
[255c4e34d10] 2025.05.11 14:13:29.538379 [ 769 ] (09b46531-elae-418d-a429-c06640829ea3) <Trace> Planner: Query from stage FetchColumns to stage Complete
[255c4e34d10] 2025.05.11 14:13:29.539507 [ 769 ] (09b46531-elae-418d-a429-c06640829ea3) <Trace> QueryPlanOptimizePrewhere: The min valid primary key position for moving to the tail of PREW
HERE is -1
[255c4e34d10] 2025.05.11 14:13:29.539542 [ 769 ] (09b46531-elae-418d-a429-c06640829ea3) <Trace> QueryPlanOptimizePrewhere: Moved 1 conditions to PREWHERE
[255c4e34d10] 2025.05.11 14:13:29.539680 [ 769 ] (09b46531-elae-418d-a429-c06640829ea3) <Debug> my_database.uk_price_paid (6e7b31aa-a3d2-4173-a918-8a6de69e2962) (SelectExecutor): Key condi
tion: unknown
[255c4e34d10] 2025.05.11 14:13:29.539719 [ 769 ] (09b46531-elae-418d-a429-c06640829ea3) <Trace> my_database.uk_price_paid (6e7b31aa-a3d2-4173-a918-8a6de69e2962) (SelectExecutor): Filtering
marks by primary and secondary keys
[255c4e34d10] 2025.05.11 14:13:29.539764 [ 769 ] (09b46531-elae-418d-a429-c06640829ea3) <Trace> QueryConditionCache: Read entry for table_uuid: 6e7b31aa-a3d2-4173-a918-8a6de69e2962, part:
all_1_1_0, condition hash: 7185031449192249839
[255c4e34d10] 2025.05.11 14:13:29.539784 [ 769 ] (09b46531-elae-418d-a429-c06640829ea3) <Debug> my_database.uk_price_paid (6e7b31aa-a3d2-4173-a918-8a6de69e2962) (SelectExecutor): Query con
dition cache has dropped 7/12 granules for PREWHERE condition equals(__table1.price, 14000 UInt16).
[255c4e34d10] 2025.05.11 14:13:29.539824 [ 769 ] (09b46531-elae-418d-a429-c06640829ea3) <Debug> QueryConditionCache: Could not find entry for table_uuid: 6e7b31aa-a3d2-4173-a918-8a6de69e29
62, part: all_1_1_0, condition hash: 933907334030584911
[255c4e34d10] 2025.05.11 14:13:29.539839 [ 769 ] (09b46531-elae-418d-a429-c06640829ea3) <Debug> my_database.uk_price_paid (6e7b31aa-a3d2-4173-a918-8a6de69e2962) (SelectExecutor): Query con
dition cache has dropped 0/5 granules for WHERE condition equals(price, 14000 UInt16).
[255c4e34d10] 2025.05.11 14:13:29.539861 [ 769 ] (09b46531-elae-418d-a429-c06640829ea3) <Debug> my_database.uk_price_paid (6e7b31aa-a3d2-4173-a918-8a6de69e2962) (SelectExecutor): Selected
1/1 parts by partition key, 1 parts by primary key, 12/12 marks by primary key, 5 marks to read from 5 ranges
[255c4e34d10] 2025.05.11 14:13:29.539882 [ 769 ] (09b46531-elae-418d-a429-c06640829ea3) <Trace> my_database.uk_price_paid (6e7b31aa-a3d2-4173-a918-8a6de69e2962) (SelectExecutor): Spreading
mark ranges among streams (default reading)
[255c4e34d10] 2025.05.11 14:13:29.539979 [ 769 ] (09b46531-elae-418d-a429-c06640829ea3) <Trace> my_database.uk_price_paid (6e7b31aa-a3d2-4173-a918-8a6de69e2962) (SelectExecutor): Reading 5
ranges in order from part all_1_1_0, approx. 40960 rows starting from 0
[255c4e34d10] 2025.05.11 14:13:29.540018 [ 769 ] (09b46531-elae-418d-a429-c06640829ea3) <Trace> MergeTreeSelectProcessor: PREWHERE condition was split into 1 steps
[255c4e34d10] 2025.05.11 14:13:29.540576 [ 1465 ] (09b46531-elae-418d-a429-c06640829ea3) <Trace> QueryConditionCache: Updated entry for table_id: 6e7b31aa-a3d2-4173-a918-8a6de69e2962, part
name: all_1_1_0, condition hash: 7185031449192249839, condition: , marks count: 13, has final mark: true
[255c4e34d10] 2025.05.11 14:13:29.553700 [ 1465 ] (09b46531-elae-418d-a429-c06640829ea3) <Trace> MergeTreeSelectProcessor: Read steps statistics: step 0 rows_read:
40960
[255c4e34d10] 2025.05.11 14:13:29.554188 [ 769 ] (09b46531-elae-418d-a429-c06640829ea3) <Debug> executeQuery: Read 40960 rows, 2.52 MiB in 0.016765 sec., 2443185.2072770656 rows/sec., 150.
08 MiB/sec.
[255c4e34d10] 2025.05.11 14:13:29.554576 [ 769 ] (09b46531-elae-418d-a429-c06640829ea3) <Debug> MemoryTracker: Query peak memory usage: 4.00 MiB.
```

Записи лога с использованием индекса – видим, что pk не найден:

(SelectExecutor): Key condition: unknown

(SelectExecutor): Filtering marks by primary and secondary keys

(SelectExecutor): Query condition cache has dropped 7/12 granules for PREWHERE condition equals(__table1.price, 14000 UInt16).

(SelectExecutor): Query condition cache has dropped 0/5 granules for WHERE condition equals(price, 14000 UInt16).

(SelectExecutor): Selected 1/1 parts by partition key, 1 parts by primary key, 12/12 marks by primary key, 5 marks to read from 5 ranges

(SelectExecutor): Spreading mark ranges among streams (default reading)

(SelectExecutor): Reading 5 ranges in order from part all_1_1_0, approx. 40960 rows starting from 0

Проверяем запрос с primary_key полем:

```
SELECT *
FROM my_database.uk_price_paid
WHERE addr2 = 'PLOT'

Query id: a82b954c-7aa9-48ec-9081-308c2400e836

[255c4e34d10] 2025.05.11 13:53:12.839536 [ 769 ] [a82b954c-7aa9-48ec-9081-308c2400e836] <Debug> executeQuery: (from 127.0.0.1:60446, user: username) (query 2, line 1) select * from my_data
base.uk_price_paid where addr2 = 'PLOT': (stage: Complete)
[255c4e34d10] 2025.05.11 13:53:12.839986 [ 769 ] [a82b954c-7aa9-48ec-9081-308c2400e836] <Trace> Planner: Query to stage Complete
[255c4e34d10] 2025.05.11 13:53:12.840180 [ 769 ] [a82b954c-7aa9-48ec-9081-308c2400e836] <Trace> Planner: Query from stage FetchColumns to stage Complete
[255c4e34d10] 2025.05.11 13:53:12.840927 [ 769 ] [a82b954c-7aa9-48ec-9081-308c2400e836] <Trace> QueryPlanOptimizePrewrite: The min valid primary key position for moving to the tail of PREW
HERE is -1
[255c4e34d10] 2025.05.11 13:53:12.840953 [ 769 ] [a82b954c-7aa9-48ec-9081-308c2400e836] <Trace> QueryPlanOptimizePrewrite: Moved 1 conditions to PREWHERE
[255c4e34d10] 2025.05.11 13:53:12.841071 [ 769 ] [a82b954c-7aa9-48ec-9081-308c2400e836] <Debug> my_database.uk_price_paid (6e7b31aa-a3d2-4173-a918-8a6de69e2962) (SelectExecutor): Key condi
tion: (column 3 in ['PLOT', 'PLOT'])
[255c4e34d10] 2025.05.11 13:53:12.841097 [ 769 ] [a82b954c-7aa9-48ec-9081-308c2400e836] <Trace> my_database.uk_price_paid (6e7b31aa-a3d2-4173-a918-8a6de69e2962) (SelectExecutor): Filtering
marks by primary and secondary keys
[255c4e34d10] 2025.05.11 13:53:12.841137 [ 769 ] [a82b954c-7aa9-48ec-9081-308c2400e836] <Trace> my_database.uk_price_paid (6e7b31aa-a3d2-4173-a918-8a6de69e2962) (SelectExecutor): Used gene
ric exclusion search over index for part all_1_1_0 with 19 steps
[255c4e34d10] 2025.05.11 13:53:12.841167 [ 769 ] [a82b954c-7aa9-48ec-9081-308c2400e836] <Trace> QueryConditionCache: Read entry for table_uuid: 6e7b31aa-a3d2-4173-a918-8a6de69e2962, part:
all_1_1_0, condition hash: 15565309393588196737
[255c4e34d10] 2025.05.11 13:53:12.841183 [ 769 ] [a82b954c-7aa9-48ec-9081-308c2400e836] <Debug> my_database.uk_price_paid (6e7b31aa-a3d2-4173-a918-8a6de69e2962) (SelectExecutor): Query con
dition cache has dropped 8/12 granules for PREWHERE condition equals( __table1.addr2, 'PLOT'_String).
[255c4e34d10] 2025.05.11 13:53:12.841201 [ 769 ] [a82b954c-7aa9-48ec-9081-308c2400e836] <Debug> QueryConditionCache: Could not find entry for table_uuid: 6e7b31aa-a3d2-4173-a918-8a6de69e29
62, part: all_1_1_0, condition hash: 14577893117438637781
[255c4e34d10] 2025.05.11 13:53:12.841218 [ 769 ] [a82b954c-7aa9-48ec-9081-308c2400e836] <Debug> my_database.uk_price_paid (6e7b31aa-a3d2-4173-a918-8a6de69e2962) (SelectExecutor): Query con
dition cache has dropped 0/4 granules for WHERE condition equals(addr2, 'PLOT'_String).
[255c4e34d10] 2025.05.11 13:53:12.841243 [ 769 ] [a82b954c-7aa9-48ec-9081-308c2400e836] <Debug> my_database.uk_price_paid (6e7b31aa-a3d2-4173-a918-8a6de69e2962) (SelectExecutor): Selected
1/1 parts by partition key, 1 parts by primary key, 12/12 marks by primary key, 4 marks to read from 3 ranges
[255c4e34d10] 2025.05.11 13:53:12.841263 [ 769 ] [a82b954c-7aa9-48ec-9081-308c2400e836] <Trace> my_database.uk_price_paid (6e7b31aa-a3d2-4173-a918-8a6de69e2962) (SelectExecutor): Spreading
mark ranges among streams (default reading)
[255c4e34d10] 2025.05.11 13:53:12.841352 [ 769 ] [a82b954c-7aa9-48ec-9081-308c2400e836] <Trace> my_database.uk_price_paid (6e7b31aa-a3d2-4173-a918-8a6de69e2962) (SelectExecutor): Reading 3
ranges in order from part all_1_1_0, approx. 32768 rows starting from 0
[255c4e34d10] 2025.05.11 13:53:12.841378 [ 769 ] [a82b954c-7aa9-48ec-9081-308c2400e836] <Trace> MergeTreeSelectProcessor: PREWHERE condition was split into 1 steps
[255c4e34d10] 2025.05.11 13:53:12.850827 [ 1444 ] [a82b954c-7aa9-48ec-9081-308c2400e836] <Trace> QueryConditionCache: Updated entry for table_id: 6e7b31aa-a3d2-4173-a918-8a6de69e2962, part
name: all_1_1_0, condition hash: 15565309393588196737, condition: , marks count: 13, has final mark: true
[255c4e34d10] 2025.05.11 13:53:12.850926 [ 1444 ] [a82b954c-7aa9-48ec-9081-308c2400e836] <Trace> MergeTreeSelectProcessor: Read steps statistics: step 0 rows_read: 32768, step 1 rows_read:
32768
[255c4e34d10] 2025.05.11 13:53:12.851971 [ 769 ] [a82b954c-7aa9-48ec-9081-308c2400e836] <Debug> executeQuery: Read 32768 rows, 2.01 MiB in 0.012456 sec., 2630700.064226076 rows/sec., 161.6
0 MiB/sec.
[255c4e34d10] 2025.05.11 13:53:12.852313 [ 769 ] [a82b954c-7aa9-48ec-9081-308c2400e836] <Debug> MemoryTracker: Query peak memory usage: 2.83 MiB.
```

Записи лога с использованием индекса – видим, что используется pk в 3м столбце:

(SelectExecutor): Key condition: (column 3 in ['PLOT', 'PLOT'])

(SelectExecutor): Filtering marks by primary and secondary keys

(SelectExecutor): Used generic exclusion search over index for part all_1_1_0 with 19 steps

(SelectExecutor): Query condition cache has dropped 8/12 granules for PREWHERE condition equals(__table1.addr2, 'PLOT'_String).

(SelectExecutor): Query condition cache has dropped 0/4 granules for WHERE condition equals(addr2, 'PLOT'_String).

(SelectExecutor): Selected 1/1 parts by partition key, 1 parts by primary key, 12/12 marks by primary key, 4 marks to read from 3 ranges

(SelectExecutor): Spreading mark ranges among streams (default reading)

(SelectExecutor): Reading 3 ranges in order from part all_1_1_0, approx. 32768 rows starting from 0

Проверяем использование индексов через EXPLAIN.

Проверяем запрос без primary_key поля:

```
EXPLAIN indexes = 1
SELECT *
FROM my_database.uk_price_paid
WHERE price = 14000

Query id: d8692e34-4fe8-4c53-af49-6e7e45e476be

explain
1. Expression ((Project names + Projection))
2. Expression
3. ReadFromMergeTree (my_database.uk_price_paid)
4. Indexes:
5. PrimaryKey
6. Condition: true
7. Parts: 1/1
8. Granules: 12/12

8 rows in set. Elapsed: 0.005 sec.
```

Полей keys в выводе нет.

Проверяем запрос с primary_key полем:

```
EXPLAIN indexes = 1
SELECT *
FROM my_database.uk_price_paid
WHERE addr2 = 'PLOT'

Query id: 5053c656-73a8-493b-a493-1d6283eb019c

explain
1. Expression ((Project names + Projection))
2. Expression
3. ReadFromMergeTree (my_database.uk_price_paid)
4. Indexes:
5. PrimaryKey
6. Keys:
7. addr2
8. Condition: (addr2 in ['PLOT', 'PLOT'])
9. Parts: 1/1
10. Granules: 12/12

10 rows in set. Elapsed: 0.007 sec.
```

Вывод показал, что используется primary_key поле addr2.